

SEAGAS -- Sprint 2 Status Report

Carl (Soh Way Miin) -- Backend Developer | 1 October 2026
COS40005 Computing Technology Project A | Group 10 | POC004

Bug Fixed Today -- Recommendations Library Encoding Issue

What was the problem?

The Recommendations page was showing duplicate resource cards for some skills. For example, under Excel, the same course 'Microsoft Excel -- Data Analysis with Excel Pivot Tables' appeared twice with slightly different garbled titles. The issue also affected Angular, React, AWS, Git, TryHackMe, Vue, FastAPI, and 20+ other skills.

Root cause:

Layer	Problem	Detail
SQL seed file	Em dash character (U+2014) in resource titles	Three original seed files used typographic em dashes (--) in titles like 'React -- The Complete Guide'
Database	54 duplicate records with corrupted titles	Two different corrupt encodings of the same title were stored: one as "Ré¼" and one as "¼"
Frontend	Both corrupted records displayed	Because the backend returned all matching records, the React frontend displayed both

How was it fixed?

Step	Action	Result
1	Diagnosed the problem via API response	GET /api/assessments/{id}/recommendations returned 'Microsoft Excel Ñ Data Analysis' and 'Microsoft Excel ¼ Data Analysis'
2	Queried database for all corrupted titles	SELECT resource_id, resource_title FROM recommendations_library WHERE resource_title LIKE 'Ré¼' OR resource_title LIKE '¼'
3	Deleted all "Ré¼" variant (27 rows)	DELETE FROM recommendations_library WHERE resource_title LIKE 'Ré¼' -- DELETE 27
4	Updated all Ñ variant to use --	UPDATE recommendations_library SET resource_title = REPLACE(resource_title, 'Ñ', '--') WHERE resource_title LIKE 'Ñ%'
5	Fixed Excel specifically (separate issue)	Deleted 2 corrupted Excel rows, inserted 1 clean row: 'Microsoft Excel - Data Analysis with Excel Pivot Tables'
6	Fixed seed file to prevent recurrence	Ran Python script to replace all em dash (U+2014) and en dash (U+2013) characters with -- in seed files
7	Pushed fix to PR #7	git commit -m 'fix: replace em dash with -- in recommendations library to prevent encoding errors'

Is it resolved?

Check	Before Fix	After Fix	Status
Duplicate resources on page	Yes (54 corrupted records)	None	FIXED
Total records in DB	326 (with 54 corrupted)	298 (all clean)	FIXED
Skills without resources	0	0	OK
Duplicate resource-skill pairs	0 (different strings)	0	OK
Seed file encoding	Contains em dash (causes errors)	Now replaced with --	FIXED
UNIQUE constraint	Active (uq_resource_skill)	Active	OK
Frontend display	Duplicate garbled cards	Clean single cards	FIXED

Sprint 2 -- What Has Been Completed (Carl)

Task	Ticket	Detail	Date
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jobs.py sort order bug	S1 carryover	Verified CASE WHEN sort is correct: required -> preferred -> Sept 30	Sept 30. No code change needed
Weighted readiness scoring	S2-05	Rewrote assessments.py: required=3pts, preferred=2pts, bonus=1pt. Formula: earned/max*100	Sept 30
Recommendations library merge	S2-03	Merged 3 SQL files (v1+v2+v3) into one seed_recommendations_library_final.sql. 326 resources	Sept 30
Remove duplicate resources	S2-03	Used ctid method to delete 174 duplicates. DB went from 478 Sept 28	Sept 28 clean records.
UNIQUE constraint	S2-03	ALTER TABLE recommendations_library ADD CONSTRAINT recommendations_source_skill UNIQUE (skill_id)	Sept 30
Fix encoding errors (54 records)	S2-03	Deleted 54 corrupted title records, re-inserted 27 clean ones Oct 1	Updated seed file to use -- instead of --
recommendation/engine.py	S2-04	Aaron's rule-based engine integrated into assessments.py. (Sept 30)	recommendations returned
PR #7 pushed	S2-03/04/05	7 commits on features/recommendation_engine -> dev. Reviewer: renhang0731, yonghin. S	

Full Team Sprint 2 Progress

Member	Role	Completed	Pending	Status
Carl (Soh Way Miin)	Backend Dev	Weighted scoring, recommendations library API (326 resources), validating Yong Hin's schema.sql UNIQUE constraint		
Ren Hang (Tee Ren Hang)	Frontend Dev	New Home page, Preloader, Login redirection, Dashboard page, new PB features, Dashboard, 401 handling	Design, Dashboard charts, PB features, Dashboard, 401 handling	review
Yong Hin (Ng Yong Hin)	AI/NLP Engineer	Skills library 32->138, job_role_skills 152 entries, 5000 students generated, 4000 jobs generated, PR pending		
Aaron (Tan Jun Xiong)	Co-Leader	recommendation/engine.py (rule-based), recommendations_library population, this sprint		report writing
Shanice (Yan Min Xuan)	Team Leader	UI/UX wireframes, Student Dashboard design	Design, Student Dashboard implementation	in progress

What Remains To Be Done

Immediate (This Week)

Priority	Task	Owner	Blocker
HIGH	Approve and merge PR #7 (features/recommendation_engine) -> dev	Ren Hang (Yong Hin)	Waiting for review approval
HIGH	Approve and merge PR feature/ui-ux -> dev	Team	Waiting for review approval
HIGH	Approve and merge PR feature/skills-library-v2 -> dev	Team	Waiting for review approval
HIGH	NLP module -- keyword matching (Level 1)	Yong Hin	Not yet started -- critical path
HIGH	NLP module -- semantic matching (Level 2, Sentence Transformers)	Yong Hin	Depends on Level 1
MEDIUM	Connect NLP module to backend API (/api/jobs/descriptions/{id}/analysis)	Carl	Waiting for Yong Hin NLP module
MEDIUM	Advisor Dashboard charts (Chart.js / Plotly)	Ren Hang	In progress
MEDIUM	Student Dashboard implementation	Shanice	In progress
LOW	Update schema.sql with UNIQUE constraint	Carl	Minor -- anytime
LOW	Collect real JD data (60 per role x 6 roles = 360 JDs)	Yong Hin	Not started

Sprint 2 Remaining Functional Requirements

FR	Requirement	Status
FR-11	Skill gap analysis against job role templates	Done (assessments.py weighted scoring)
FR-12	Personalised learning recommendations per skill gap	Done (engine.py + 298 resources)
FR-13	Student readiness score (7 dimensions)	Done (assessments.py dimension scores)
FR-14	Level 1 keyword matching vs Level 2 semantic matching	Level 1 done -- Level 2 pending (Yong Hin NLP)
FR-15	Advisor dashboard -- cohort overview and analytics	In progress (Ren Hang)
FR-16	Student dashboard -- personal progress view	In progress (Shanice)
FR-17	JD upload and NLP skill extraction	Backend endpoint ready -- NLP integration pending
FR-18	Job description analysis (real JD data)	Not started -- 60 JDs per role needed

Biggest Risk

Yong Hin's NLP module has not been started yet. This is the core AI component of SEAGAS and is on the critical path -- Carl's NLP API integration, FR-14 (Level 2 semantic matching), and FR-17 (JD skill extraction) all depend on it. If this is not completed by mid-Sprint 2, the system will only have placeholder keyword matching for the demo.

Open Pull Requests

PR	Branch	Author	Contents	Reviewers	Status
#7	features/recommendation_engine -> dev	Carl	Weighted scoring, 298 resources, encoding 0731	Yong Hin, Shanice	Pending approval
#?	feature/ui-ux -> dev	Ren Hang	New UI (Home, Login, Register, Logout) Brand colours, 40% done	Carl	Pending review
#?	feature/skills-library-v2 -> dev	Yong Hin	Skills library 138 skills, 152 job_role_TSDs, 5000 students ready to review	Carl	Pending review